This document has been approved for public release and sale; its distribution is unlimited.

MIL-S-24412(SHIPS)
1 July 1970
(See 6.4)

MILITARY SPECIFICATION

STEEL, SPECIAL STRUCTURAL SHAPE, WELDABLE, HIGH TENSILE (HT); FOR SUBMARINE HULLS

1. SCOPE

1.1 Scope. This specification covers one size high tensile steel structural shape for use in existing submarine hulls for over haul and maintenance purposes.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

FEDERAL
TT-P-645 - Primer, Paint, Zinc-Chromate, Alkyd Type.
TT-P-664 - Primer, Coating, Synthetic, Rust Inhibiting, Lacquer Resisting.

MILITARY
MIL-P-8585 - Primer Coating, Zinc-Chromate, Low Moisture Sensitivity.
MIL-P-15328 - Primer (Wash), Pretreatment, Blue (Formula No. 117B for Metals).
MIL-P-15930 - Primer, Coating, Shipboard, Vinyl-Zinc Chromate (Formula No. 120 - For Hot Spray).

STANDARDS

FEDERAL
FED-STD-66 - Steel; Chemical Composition and Hardenability.
FED-STD-151 - Metals; Test Methods.
FED-STD-595 - Colors.

MILITARY
MIL-STD-163 - Steel Mill Products, Preparation for Shipment and Storage.
MIL-STD-248 - Welding and Brazing Procedure and Performance Qualification.
MIL-STD-271 - Non Destructive Testing Requirements for Metals.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions, should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
A-6 - General Requirements for Delivery of Rolled Steel Plates, Shapes, Sheet Piling and Bar for Structural Use.
E-208 - Method for Conducting Drop-Weight Tests to Determine Nil-Ductility Transition Temperature of Ferritic Steels.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

FSC 9520

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:51Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Sample for first article inspection. Prior to beginning production a sample shall be tested as specified in 4.2 (see 6.3).

3.2 Material. The shape shall be produced from basic open hearth, basic oxygen or electric furnace ingots, blooms or billets. The steel shall be made to fully killed, fine grain practice.

3.3 Chemical composition. The chemical composition of the steel, as determined by the ladle analysis, shall conform to table I.

Table I. Chemical composition (ladle analysis)[1/] [2/]

Carbon max.	Manganese max.	Phosphor max.	Sulfur max.	Silicon max.	Vanadium min.
Percent	Percent	Percent	Percent	Percent	Percent
0.18	1.30	.035	.040	.15/.35	.02

1/ All other elements shall be residuals as melted.

2/ Check analysis tolerances shall be in accordance with the applicable table in FED-STD-66.

3.4 Mechanical properties. The mechanical properties shall conform to table II.

Table II. Mechanical properties.

Yield strength[1/]	Ultimate tensile strength	Elongation min.
		8 in. gage
Min. psi	psi	percent
45,000	90,000 max.	20

1/ Yield strength (.2 percent offset).

3.5 Dimensions. Dimensions shall be in accordance with figure 1.

3.5.1 Tolerances. All tolerances not specified in figure 1, shall meet the weight and dimensional requirements of ASTM A-6.

3.6 Internal soundness.

3.6.1 Each shape shall be ultrasonically tested and meet the following acceptance standards in accordance with 4.5.3. When scanning, complete loss or transposition of back reflection over a 4 square inch surface area shall be cause for rejection of the shape. Discontinuities resulting in 50 percent or greater loss of back reflection shall be recorded and the adjacent surface within six inches in the same plane shall be continuously scanned. If two or more discontinuities resulting in 50 percent or greater loss of back reflection occur within six inches of each other, they shall be further examined with a 3/4-inch diameter (maximum) crystal. Individual discontinuities must be separated by 1/8-inch (minimum) in the thickness dimension or one ultrasonic crystal diameter (3/4-inch) to be considered as being separate. Discontinuities which do not meet the separation criteria above are to be considered as one discontinuity. Complete loss or transposition of back reflection due to a discontinuity with a total area over four square inches, shall be cause for rejection.

3.7 Surface soundness.

3.7.1 All structural shapes shall meet the requirements of ASTM A-6.

3.7.2 Surface defects shall be repaired in accordance with ASTM A-6 with the following additional requirements.

(a) Repair welding shall be carried out employing welders qualified to MIL-STD-248.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:51Z
Check the source to verify that this is the current version before use.

(b) Repair welds shall be carried out employing a welding procedure that has been qualified in accordance with MIL-STD-248.
(c) Repair welds shall be magnetic particle tested after welds are ground flush with the surface. Magnetic particle testing shall be in accordance with MIL-STD-271. No linear or linear disposed indications are permitted.

3.7.3 Structural shapes produced to this specification shall have a surface suitable for ultrasonic inspection in areas specified for scanning.

3.8 Cleaning and preservation of shape surfaces.

3.8.1 Each shape shall be furnished descaled and coated.

3.8.1.1 Shapes shall be descaled and cleaned by abrasive blast cleaning or acid pickling.

3.8.1.2 Abrasive blast cleaning. Abrasive blast cleaning shall result in a clean metal surface, suitable for painting, with mill scale, rust, and other surface contaminants completely removed.

3.8.1.3 Acid pickling. The acid pickling process shall result in clean metal surface suitable for painting, with scale, rust, and other surface contaminants completely removed.

3.8.2 Coating. The paint film shall be sufficient to cover surface roughness peaks adequately. Two random dry film thickness measurements per 100 square feet of painted surface, made with a suitable thickness gage, shall be considered sufficient for determining conformity of any one shape to the specified coating thicknesses. Other methods of measurement or quality control may be used for paint film thickness, subject to the approval of the Command or agency concerned. The shapes shall be coated dark green in accordance with (a), or (b) below:

(a) One coat of pretreatment coating, formula NO. 117B, of MIL-P-15328, to an approximate dry film thickness of 0.3 to 0.5 mils:
Followed by one coat of vinyl zinc chromate primer, formula No. 120 of MIL-P-15930, modified by the replacement of lamp black and magnesium silicate, as necessary, with sufficient and suitable insoluble inorganic pigments to produce a dark green finish, matching formula 84D/47 (FED-STD-595, color No. 34128) dry film thickness for formula No. 120, approximately 1.5 to 2.0 mils. The total dry film thickness of formula No. 117B and formula No. 120 shall be approximately 2 mils and the total thickness at any one point shall not be less than 1.5 mils.
(b) One coat of alkyd primer, formula 84D (dark green), conforming to TT-P-645 modified to a color matching formula 84D/47 (FED-STD-595, color No. 34128) to an average dry film thickness of approximately one mil. The thickness of the dry film shall not be less than 0.7 mils at any point. The choice of coating should depend on the end use (see 3.8.2.4).

3.8.2.1 The drying time of the coatings specified in 3.8.2 at 73°F. shall be as follows:

Specification	Type	Drying time maximum
MIL-P-15328	Formula 117B	30 minutes
MIL-P-15930	Formula 120	30 minutes
TT-P-645	Formula 84D	6 hours

Higher shape temperatures or ambient temperature would shorten the drying time.

3.8.2.2 The following primers are acceptable in lieu of TT-P-645, formula 84D, when modified to meet the color requirements of 3.8.2. These offer the desired reduction in drying time for effective use of automated cleaning and painting. They are subject to the same limitations as formula 84D specified in 3.8.2.4.

Specification	Drying time maximum
MIL-P-8585	30 minutes
TT-P-664	20 minutes

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:51Z
Check the source to verify that this is the current version before use.

3.8.2.3 A dry color chip for primer formula 84D/47 may be obtained from General Services Administration, Specifications Activities, Materials Division, Regional Office Building, 7th and D Streets, S.W., Washington, D.C. 20407.

3.8.2.4 The coatings specified herein should be selected in terms of the particular use. The vinyl primer dries much more rapidly than the alkyd primer and provides a longer term protection. Cleaning, followed by one coat of alkyd primer, formula 84D of TT-P-645, modified TT-P-664 or MIL-P-8585, should provide protection for approximately 9 months. Cleaning, followed by one coat of the pretreatment coating, formula 117B of MIL-P-15328, and one coat of vinyl primer, formula 120 of MIL-P-15930 should provide protection for approximately two years. Consideration should be given to fabrication problems in cutting and welding shapes coated with vinyl paint systems.

3.9 Identification marking. Each shape shall be marked near one end with the specification number, grade, ("HT"), size of section, and manufacturer's name or trademark. Shapes that have passed the ultrasonic tests specified herein, shall be marked "U.T.". The marking shall be either indent stamped or painted on the piece with a greaseless waterproof paint.

3.10 Documentation. A copy of the mill certificates and inspection data shall be sent to the consignee.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection. First article inspection shall consist of the examination and tests specified in 4.7.

4.3 Sampling for quality conformance.

4.3.1 Lot definition.

4.3.1.1 A lot for the purposes of chemical analysis and mechanical properties shall be defined as the product from a single heat or melt.

4.3.1.1.1 From each lot, one chemical analysis (ladle analysis) shall be performed.

4.3.2 Mechanical properties. From each lot, two tensile tests shall be performed. If a lot consists of ten shapes or less, one tensile test shall suffice.

4.3.3 Visual and dimensional examination. Shapes shall be visually and dimensionally examined in accordance with ASTM A-6.

4.4 Location and orientation of test samples.

4.4.1 Tension tests shall be oriented longitudinal to the length of the shape in the web of the shape.

4.5 Tests.

4.5.1 Chemical analysis. Chemical analysis shall be performed in accordance with methods 111.2 or 112.2 of FED-STD-151.

4.5.2 Tension test. Tension tests shall be performed in accordance with ASTM A-6.

4.5.3 Ultrasonic test. Each shape shall be ultrasonically tested by a procedure qualified in accordance with the requirements specified for plates in MIL-STD-271. The procedure shall employ a minimum frequency of 2.25 MHz with a maximum diameter crystal of 1.125 inches. The beams shall be tested as follows: A minimum of three longitudinal, continuous scans on the web; one along the center of the web and one each along the two web-flange intersections as close to the web-flange radius as practical.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:51Z
Check the source to verify that this is the current version before use.

4.6 Retest provisions.

4.6.1 Mechanical testing. Lots rejected due to tensile test failure may be reinstated by tensile testing each shape and determining that it meets the tensile requirements of this specification. When material is identifiable as to the ingot and cut, it will suffice to blanket acceptable material by tension testing the extremes of an ingot or group of ingots. For example, consider the shapes representing ingots 7, 8, and 9 to have failed in the original tensile lot. Tension testing the top of the top shape in ingot number 7 and the bottom of the bottom shape in ingot number 9 will reinstate all shapes produced between ingots 7 and 9.

4.7 First article tests. Prior to delivering shapes to this specification, prospective producers shall demonstrate that their facilities are capable of producing acceptable quality shapes.

4.7.1 First article sample. The first article sample shall consist of steel shape rolled from a portion of product of one heat or melt and shall be representative of normal production.

4.7.2 For procurements, the following first article tests shall be performed:

(a) Chemical composition. A ladle analysis and check analysis, respectively, shall be made of the heat or melt of steel and the rolled product involved in the first article test and shall comply with Table I. To settle disputes over chemical analysis, the FED-STD-66 standard tolerances for carbon steels shall apply.

(b) Mechanical tests. Two longitudinal flat tension test specimens shall be taken from the web of the first article sample and shall comply with table II.

(c) Toughness tests. A charpy vee notch curve, using sub size specimens, which will show the upper and lower energy shelves, is required. The material shall show a nil ductility transition (NDT) temperature of 0°F. by change of slope method. As an alternative, the following methods may be used utilizing material from the same heat and rolled to sections 1 inch or less in thickness:

(1) Drop weight test (ASTM E-208) with an NDT temperature of 0°F. required.

(2) Full size charpy vee notch specimens showing the full transition curve. The material shall show a NDT temperature of 0°F. by the change of slope method.

4.7.3 The tests and records of the tests shall be verified by the cognizant Government representative. If practicable,the Government representative should witness the tests. When a producer has more than one plant, first article test data is required for each plant. This does not apply to facilities at the same location which employ identical processing procedures. Any form of reporting the test results which gives the pertinent information and is satisfactory to the Government representative may be used. The report should include sketches of the shapes showing locations of the test specimens. The requirements for reporting results given in FED-STD-151 do not apply. The test report should be forwarded to the Commander, Naval Ship Engineering Center, Materials Development and Applications Office via the Government representative.

4.8 Inspection for preparation for delivery. An examination shall be made to determine that preservation, packaging, packing and marking comply with the requirements of section 5 of this specification.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements for the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.5.)

5.1 Preservation, packaging, packing and marking. Structural shapes shall be packaged, packed and marked level A or C as specified (see 6.1) in accordance with MIL-STD-163. Preservation shall be in accordance with 3.8.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:51Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Length and number of shapes required.
(c) Level of packaging, packing and marking required (see 5.1).

6.2 When shapes are procured under this specification from warehouses or sources other than producers, the suppliers shall be required to furnish verified test data to the effect that each lot of steel has been produced by a satisfactory procedure established in accordance with the first article test requirements of this specification.

6.3 First article testing.

6.3.1 Invitations for bids should provide that the Government reserves the right to waive the requirement for first article test samples as to those bidders offering this size shape which has been previously procured or tested by the Government, under MIL-S-20166, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.3.2 When a major change in the manufacturing process occurs, the Government reserves the right to require additional first article testing.

6.4 Superseding data. This specification replaces one size of grade HT high tensile steel of MIL-S-20166A. Shapes previously procured under MIL-S-20166A to FSN 9520-277-7299 and meeting UT standards and cleaned and preserved are acceptable for submarine service and may be used until supply is exhausted.

6.5 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in Section 2 do not apply when material and parts are procured by the supplier for incorporation into the equipment and lose their separate identity when the equipment is shipped.

Preparing activity:
Navy - SH
(Project 9520-N001)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:51Z
Check the source to verify that this is the current version before use.

SPECIAL H-BEAM DIMENSION TOLERANCES

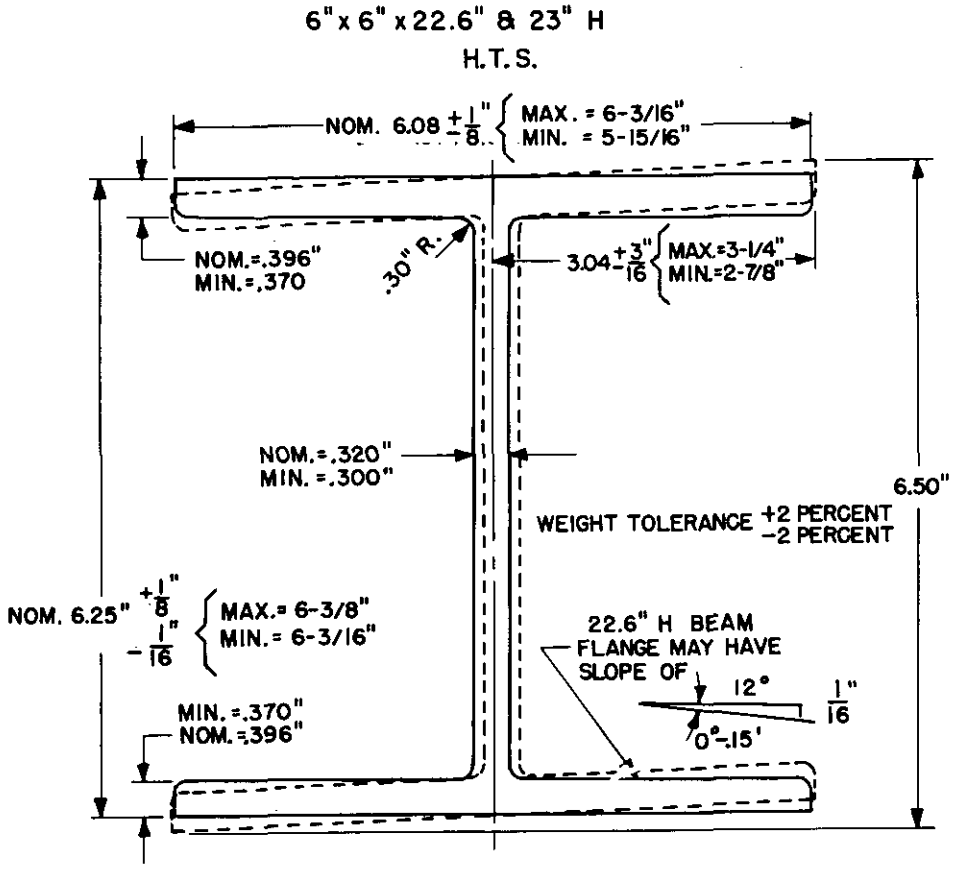


Figure 1 - Dimension tolerances for special structural shape steel.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:51Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY
Naval Ship Engineering Center
Center Building
Prince George's Center
Hyattsville, Maryland 20782

OFFICIAL BUSINESS

POSTAGE AND FEES PAID
NAVY DEPARTMENT

Commander, Naval Ship Engineering Center
DOD Standardization Program & Documents Branch
Center Building
Prince George's Center
Hyattsville, Maryland 20782

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:51Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE 0101-807-2000

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T21:51Z
Check the source to verify that this is the current version before use.